



STRAT*EX*

PROJECT AD-KY041 • 142 Bluegrass Ridge • Lexington, KY

CONFIDENTIAL • STRATEGIC FORENSIC AUDIT



STRAT*EX*TM

FORENSIC ENVELOPE AUDIT • V1.0

Strategic Thermal Reconnaissance Report

142 Bluegrass Ridge • Lexington, KY • Scan 2026-06-25 • Ground-truth accuracy ±0.78 cm

TOTAL SQUARES

24.31

SQ (100 sf)

SHINGLE LAYERS

2 / 2

DETECTED / CODE MAX

FACETS

11

DISTINCT PLANES

ANOMALIES FLAGGED

3

FORENSIC HOTSPOTS

EXECUTIVE SUMMARY • GRAND TOTAL RANGE

\$32,491.38 → \$33,996.38

MATERIALS \$23,407.50 • LABOR \$6,028.88 • TEAR-OFF \$1,850.00 • UNFORESEEN \$1,205.00-\$2,710.00

STRATEX RECOMMENDATION

TEAR OFF

Two existing shingle layers + 88% sub-surface moisture probability exceeds code re-roof limit. Full tear-off, decking remediation, and new-install is the only insurable path.

• FOR YOU • THE HOMEOWNER

What We Found, In Plain English

YOUR PROPERTY • BIG PICTURE

- Your roof was scanned with a drone — every measurement is accurate to less than a centimeter.
- We found **3 problem area(s)** that need attention.
- Your home’s overall envelope score is **55/100**.
- We strongly recommend a **complete tear-off and re-roof**.
- You are losing about **\$1,095.00/year** through air leaks.

WHAT THIS WILL COST

\$32,491.38 – \$33,996.38

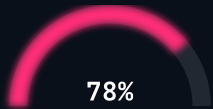
ALL-IN • MATERIALS + LABOR + TEAR-OFF + RESERVE

PRIORITY • DO THESE FIRST

URGENT	Replace Front Door Weatherstripping
HIGH	Remediate Blocked Attic Soffit Vents
HIGH	Inspect and Seal 2nd-Story Window Headers

Building Envelope Score

ENERGY DEFICIENCY



DEFICIENT

VENTILATION



POOR

OVERALL ENVELOPE

55/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

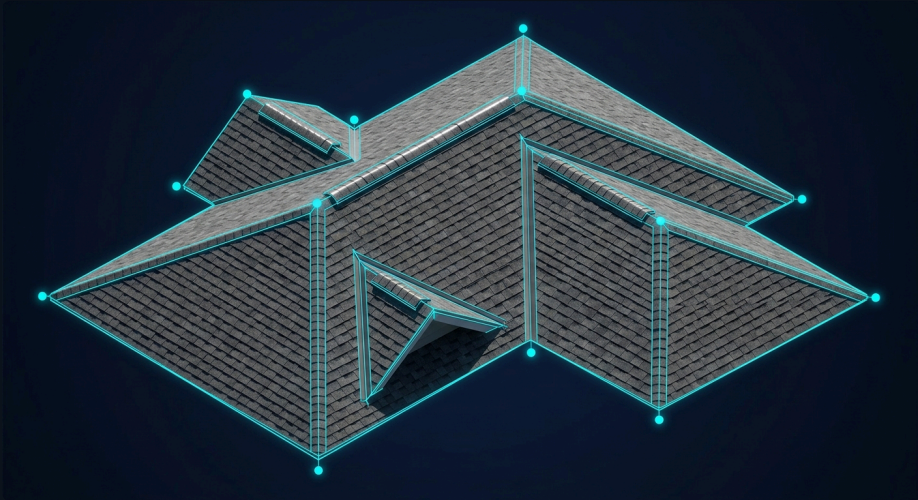
1	URGENT Replace Front Door Weatherstripping	\$150/yr
2	HIGH Remediate Blocked Attic Soffit Vents	\$110/yr
3	HIGH Inspect and Seal 2nd-Story Window Headers	\$95/yr
4	MED Add Attic Baffles for Cross-Ventilation	\$70/yr

THERMAL FORENSIC FINDINGS

Ridge Vent Performance	LOW FLOW	MED
Soffit Vents Open	8 / 22	HIGH
Front Door Seal Failure	SEVERE	SEVERE
2nd-Story Window Headers	WARM (+4.1°C)	HIGH

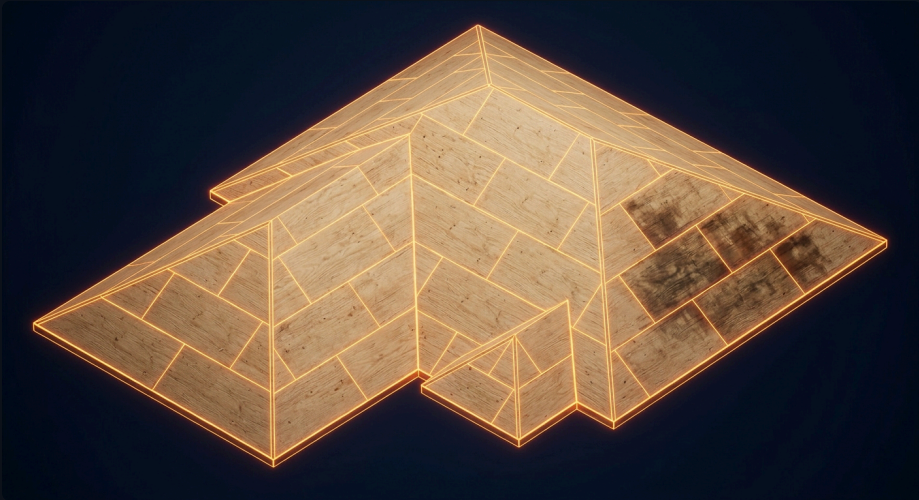
Layered CAD/BIM Twin

LAYER 01 • FINISHED SLATE



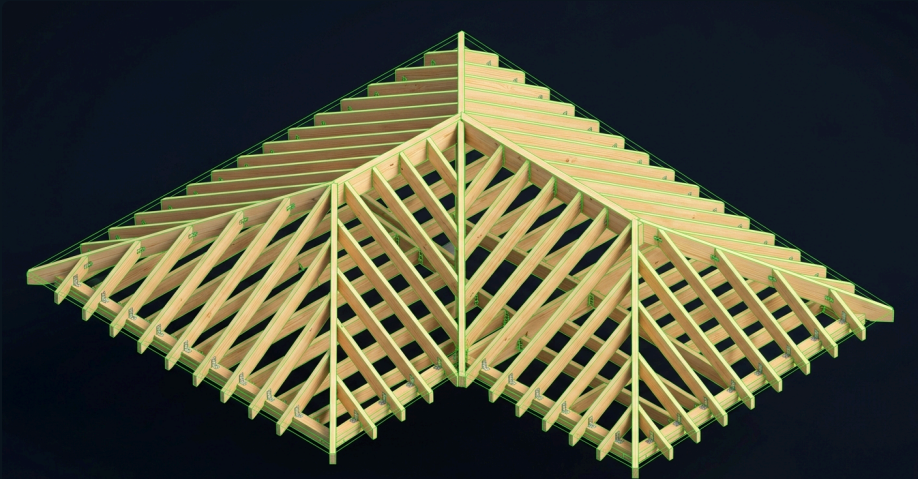
Squares 24.31 • Pitch 8/12

LAYER 02 • DECKING + UNDERLAYMENT



7/16" CDX • I&W shield at valleys/eaves • 2 existing shingle layer(s)

LAYER 03 • STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. • LVL ridge • Hurricane ties **VERIFIED**

VALLEYS

148.7

LINEAR FT

GABLES

192.3

LINEAR FT

RIDGES

62.0

LINEAR FT

EAVES

188.4

LINEAR FT

Crew Choreography

PHASE	CREW	DURATION	TIMELINE • 2.7 d
Mobilization & Site Setup	Supervision	0.25 d	
Tear-Off (existing 2 layers)	Roof Crew	0.50 d	
Decking Remediation (CDX rot)	Carpentry	0.50 d	
Underlayment + I&W Shield	Roof Crew	0.40 d	
Shingle Install (24.3 sq)	Roof Crew	0.70 d	
Wall Cavity Vapor Repair	Carpentry	1.00 d	
Hardie Siding Hang + Trim	Sheet-Metal	0.90 d	
Gutter + Downspout Install	Sheet-Metal	0.20 d	
Weatherstrip / Energy Sealing	Carpentry	0.10 d	
Cleanup + Final Inspection	Supervision	0.10 d	

Final Project Summary

TOTAL PROJECT COST

\$32,491.38

LOW • GRAND TOTAL

HIGH RANGE

\$33,996.38

HIGH • GRAND TOTAL

PROJECTED NET PROFIT

\$9,142.07

20% TARGET

URGENCY ITEMS

3

CRITICAL TASKS

AI-Prioritized Repair Plan			
#	URGENCY	TASK	EST. ANNUAL SAVINGS
1	URGENT	Replace Front Door Weatherstripping	\$150/yr
2	HIGH	Remediate Blocked Attic Soffit Vents	\$110/yr
3	HIGH	Inspect and Seal 2nd-Story Window Headers	\$95/yr
4	MED	Add Attic Baffles for Cross-Ventilation	\$70/yr

CERTIFICATION

I certify the foregoing measurements, photographic evidence, thermal radiometry, and damage classifications were captured under STRATEX™ Phase-3 Expert Agent protocols with ground-truth accuracy of ±0.78 cm. This report is suitable for insurance carrier review.

INSPECTOR

STRATEX™ Forensic Division • A. Cross, RRO

KY-RR0-08821 • NRCIA-CRI-7741

REPORT ID

AD-KY041

ISSUED 2026-06-25

DISCLAIMER • This report is the proprietary work product of STRATEX™ Forensic Division. Findings are based on aerial drone reconnaissance, photogrammetric mesh reconstruction, and radiometric thermal analysis captured on the scan date. Concealed conditions discoverable only after tear-off are documented in the unforeseen-repair reserve. Reproduction or transmission without authorization is prohibited.